

CricSense

API Documentation

v1.0 · REST API · FastAPI · Node.js / Express

Base URLs

<https://cricsense.vercel.app/>

Auth

Bearer JWT token
(Authorization header)

Content-Type

application/json
multipart/form-data (upload)

Services

Backend: <https://cricsense-ytl3.onrender.com/>
Ai: <https://cricsense-ai-slvx.onrender.com/>

1. Overview

CricSense exposes two HTTP APIs. The Backend API (Node.js/Express, port 5000) handles auth, users and analysis records. The AI API (Python/FastAPI, port 8000) is internal — called by the backend only, never by the browser.

1.1 Error Format

All errors return JSON with a message field:

```
{
  "message": "Invalid credentials"
}
```

Status	Meaning
200	Success
201	Created
400	Bad request
401	Unauthorized — invalid or missing JWT
403	Forbidden
404	Not found
500	Server error

2. Auth Routes

Base path: `/api/auth` · No authentication required

2.1 Register

POST	<code>/api/auth/register</code>	Create a new user account
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Request Body

Field	Type	Required	Description
name	string	Yes	Full name
email	string	Yes	Unique email address
password	string	Yes	Minimum 6 characters

role	string	No	player (default) or coach
------	--------	----	---------------------------

Request Example

```
POST /api/auth/register
Content-Type: application/json
```

```
{
  "name": "Virat Kohli",
  "email": "virat@cricsense.com",
  "password": "secret123",
  "role": "player"
}
```

Response 201

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": {
    "id": "664f1a2b3c4d5e6f7a8b9c0d",
    "name": "Virat Kohli",
    "email": "virat@cricsense.com",
    "role": "player"
  }
}
```

2.2 Login

POST	/api/auth/login	Authenticate and receive a JWT
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Request Body

Field	Type	Required	Description
email	string	Yes	Registered email
password	string	Yes	Account password

Request Example

```
POST /api/auth/login
Content-Type: application/json
```

```
{
  "email": "virat@cricsense.com",
  "password": "secret123"
}
```

Response 200

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": { "id": "...", "name": "Virat Kohli", "role": "player" }
}
```

2.3 Get Current User

GET	/api/auth/me	Return the authenticated user
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Response 200

```
{
  "user": {
    "_id": "664f1a2b...",
    "name": "Virat Kohli",
    "email": "virat@cricsense.com",
    "role": "player",
    "createdAt": "2024-06-01T10:30:00.000Z"
  }
}
```

3. Analysis Routes

Base path: /api/analysis · All routes require Bearer JWT

3.1 Upload Video

POST	/api/analysis/upload	Upload video and start analysis
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Returns immediately with an analysisId — processing is async. Poll GET /api/analysis/:id to check when status changes from processing to done.

Form Fields

Field	Type	Required	Description
video	File	Yes	MP4 / MOV / AVI — max 500 MB
mode	string	Yes	bat or bowl
title	string	No	Human-readable label

cURL Example

```
curl -X POST http://localhost:5000/api/analysis/upload \  
-H "Authorization: Bearer <token>" \  
-F "video=@batting.mp4" \  
-F "mode=bat" \  
-F "title=Practice June 2024"
```

Response 200

```
{  
  "analysisId": "664f1a2b3c4d5e6f7a8b9c0e",  
  "message": "Processing started"  
}
```

3.2 Get Analysis

GET	/api/analysis/:id	Fetch a single analysis with all data
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Response 200 — status: done (batting)

```
{  
  "_id": "664f1a2b3c4d5e6f7a8b9c0e",  
  "status": "done",  
  "mode": "bat",  
  "summary": {  
    "total_frames": 450, "duration_s": 15.0,  
    "right_leg_pct": 42.2, "left_leg_pct": 31.1, "balanced_pct": 26.7,  
    "avg_r_knee": 138.4, "avg_l_knee": 141.2,  
    "avg_hip_rot": 22.6, "max_offset": 68.3  
  },  
  "frames": [  
    { "frame": 1, "time_s": 0.033, "weight_side": "BALANCED",  
      "hip_offset_px": 12.4, "r_knee_angle": 140.2,  
      "l_knee_angle": 142.1, "hip_rotation": 18.5 },  
    ...  
  ]  
}
```

Response 200 — status: done (bowling)

```
{  
  "status": "done",  
  "mode": "bowl",  
  "summary": {  
    "total_frames": 360, "duration_s": 12.0,  
    "avg_arm_angle": 162.3, "avg_hip_sho_sep": 28.4,  
    "avg_front_knee": 158.9, "avg_trunk_lean": 22.1  
  }  
}
```

```

    },
    "frames": [
      { "frame": 1, "time_s": 0.033, "weight_side": "RIGHT LEG",
        "r_arm_angle": 165.4, "hip_sho_separation": 29.3,
        "front_knee_l": 162.0, "trunk_lean": 24.5, "trunk_direction": "FORWARD" }
    ]
  }
}

```

3.3 List Analyses

GET	<code>/api/analysis</code>	List current user's analyses (no frames field)
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Response 200

```

[
  { "_id": "...", "mode": "bat", "title": "Practice June 2024",
    "status": "done", "summary": { ... }, "createdAt": "2024-06-01T10:30:00.000Z"
  },
  ...
]

```

3.4 Delete Analysis

DELETE	<code>/api/analysis/:id</code>	Delete an analysis record
---------------	--------------------------------	---------------------------

Response 200

```

{
  "message": "Deleted"
}

```

4. User Routes

Base path: `/api/user` · All routes require Bearer JWT

4.1 Get Profile

GET	<code>/api/user/profile</code>	Return the current user profile
------------	--------------------------------	---------------------------------

Response 200

```

{

```

```
"user": { "_id": "...", "name": "Virat Kohli", "email": "virat@cricsense.com",  
"role": "player" }  
}
```

4.2 Update Profile

PUT	/api/user/profile	Update name or role
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Request Body

Field	Type	Required	Description
name	string	No	New display name
role	string	No	player or coach

Response 200

```
{  
  "user": { "_id": "...", "name": "Virat Kohli", "role": "coach" }  
}
```

5. AI Service API

Base URL: <http://localhost:8000> · Internal — called by the backend, not the browser

5.1 Health Check

GET	/health	Confirm AI service is running
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Response 200

```
{  
  "status": "ok"  
}
```

5.2 Analyze Video

POST	/analyze	Process video, return metrics
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Accepts multipart/form-data. Runs MediaPipe pose detection per frame. Returns summary metrics and first 50 frame rows.

Form Fields

Field	Type	Required	Description
video	File	Yes	Video file
mode	string	Yes	bat or bowl

Response 200

```
{
  "status": "success",
  "mode": "bat",
  "video_out": "/tmp/abc_analyzed.mp4",
  "csv_path": "/tmp/abc_analyzed_data.csv",
  "metrics": { "total_frames": 450, "avg_r_knee": 138.4, ... },
  "frames": [ { "frame": 1, "time_s": 0.033, ... }, ... ]
}
```

6. Data Schemas

6.1 User

Field	Type	Description
_id	ObjectId	Auto-generated MongoDB ID
name	string	Full name
email	string	Unique email (lowercase)
password	string	Bcrypt-hashed — never returned
role	string	player or coach
createdAt	Date	ISO 8601 timestamp

6.2 Analysis

Field	Type	Description
_id	ObjectId	Auto-generated ID
user	ObjectId	Ref to User
mode	string	bat or bowl
title	string	Human-readable label
status	string	processing done failed
summary	object	Averaged metrics (see 6.3 / 6.4)

frames	array	Per-frame metric rows
videoUrl	string	Path to annotated video
csvUrl	string	Path to CSV export
createdAt	Date	ISO 8601 timestamp

6.3 Batting Frame Row

Field	Type	Description
frame	int	Frame number (1-indexed)
time_s	float	Timestamp in seconds
weight_side	string	RIGHT LEG LEFT LEG BALANCED UNKNOWN
hip_offset_px	float	Hip midpoint x minus ankle midpoint x (pixels)
r_knee_angle	float null	Right knee angle in degrees
l_knee_angle	float null	Left knee angle in degrees
hip_rotation	float null	Hip line angle from horizontal in degrees

6.4 Bowling Frame Row

Field	Type	Description
frame	int	Frame number
time_s	float	Timestamp in seconds
weight_side	string	RIGHT LEG LEFT LEG BALANCED UNKNOWN
hip_offset_px	float	Hip midpoint x minus ankle midpoint x
r_arm_angle	float null	Right elbow angle in degrees
l_arm_angle	float null	Left elbow angle in degrees
hip_sho_separation	float null	Hip angle minus shoulder angle (abs value)
front_knee_l	float null	Left knee angle in degrees
trunk_lean	float null	Trunk lean angle from vertical
trunk_direction	string	FORWARD UPRIGHT BACK UNKNOWN

7. Benchmark Reference

7.1 Batting

Metric	Ideal Range	Buffer	Alert if
Right Knee Angle	120°–160°	±15°	< 105° or > 175°
Left Knee Angle	120°–160°	±15°	< 105° or > 175°
Hip Rotation	10°–35°	±8°	< 2° or > 43°
Hip Offset	-40–+40 px	±20 px	< -60 px or > +60 px

7.2 Bowling

Metric	Ideal Range	Buffer	Alert if
Bowling Arm Angle	140°–180°	±15°	< 125°
Hip-Shoulder Separation	15°–45°	±8°	< 7° or > 53°
Front Knee Bracing	150°–180°	±15°	< 135°
Trunk Lean	10°–40°	±8°	< 2° or > 48°

8. Quick Reference

8.1 All Endpoints

Method	Path	Auth	Description
POST	/api/auth/register	No	Create account
POST	/api/auth/login	No	Log in, get JWT
GET	/api/auth/me	Yes	Current user
GET	/api/user/profile	Yes	Get profile
PUT	/api/user/profile	Yes	Update profile
POST	/api/analysis/upload	Yes	Upload video
GET	/api/analysis	Yes	List analyses
GET	/api/analysis/:id	Yes	Get one analysis
DELETE	/api/analysis/:id	Yes	Delete analysis
GET	/health	No	AI health check
POST	/analyze	No	AI: process video (internal)

8.2 Environment Variables

Variable	Service	Default	Description
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PORT	Backend	5000	Server port
MONGO_URI	Backend	mongodb://localhost:27017/cricsense	MongoDB connection string
JWT_SECRET	Backend	(required)	Secret for signing JWTs
JWT_EXPIRE	Backend	7d	Token expiry
AI_API_URL	Backend	https://cricsense-ai-slvx.onrender.com/	AI service base URL
FRONTEND_URL	Backend	https://cricsense.vercel.app/	CORS allowed origin
NEXT_PUBLIC_API_URL	Frontend	https://cricsense-ytl3.onrender.com	Backend API base URL